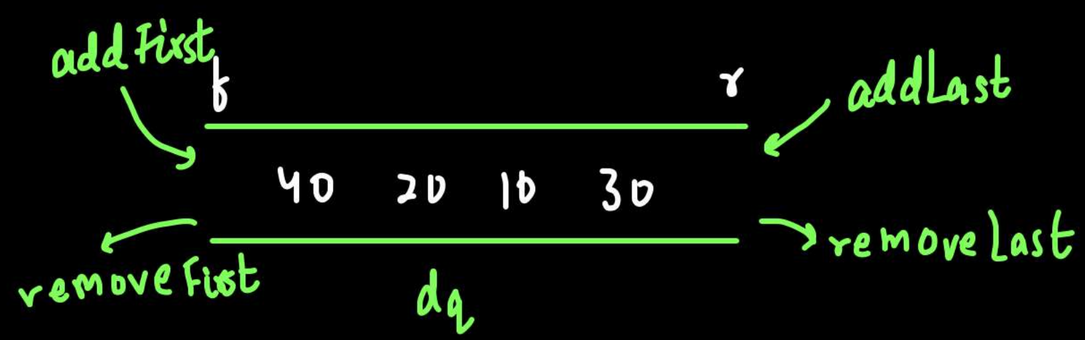
Deque (doubly ended queue) (Queue + Stack)



<u>Homework</u> : Implement these methods in a deque.

a) add(idx)

b) get(idx)

c) remove(idx)

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Subarray

Ques: Sliding Window Maximum

arr[] = [1, 2, 3, 1, 4, 5, 2, 3, 4], k = 3

total windows = n-(k-1)

Method-1 : Brute Force

T.C. = $O(n*k)$

$\cong O(n^2)$

Method-3: Deque (monotonically decreasing first to last)

b l

~~1~~ ~~2~~ ~~3~~ ~~1~~ ~~4~~ ~~5~~ ~~2~~ ~~3~~ 4

dq

idx

steps

while(out of window)
dq.removeFirst

while(dq.last < arr[i])
dq.removeLast

mark ans → dq.first

Method-2 : Next Greater Element

but..

{1, 2, 3, 4, 5, 6, 4, 8}

T.C. = $O(n*k)$

↓

Can be improved to $O(n)$

	0	1	2	3	4	5	6
ans =	3	3	4	5	5	5	4

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Ques: Sliding Window Maximum

Steps

```
while(out of window(dq.first))
    dq.removeFirst

while(dq.last < arr[i])
    dq.removeLast
dq.addLast(arr[i])
mark ans → dq.first
```

$i-k+1$ i

arr = 7 6 5 1 2 3 4 , k=3

f l

~~7~~ ~~6~~ ~~5~~ ~~1~~ ~~2~~ ~~3~~ 4

dq

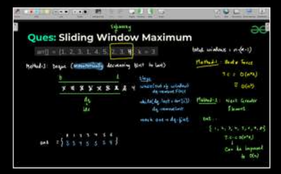
dq.first $\leq i-k$

↓

out of window

0	1	2	3	4
7	6	5	3	4

ans = { 7 6 5 3 4 }



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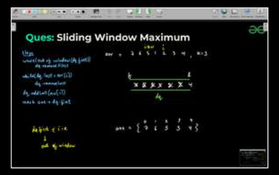
Stack nge wala method

arr = { 1, 2, 3, 4, 5, 6, 7, 8 } k = 5

nge = { 2 3 4 5 6 7 8 -1 }

ans = { 5 6

T.C. ≈ O(n)

(Code : HW)